

LAB 02

Main Functional Requirements & Use Case Diagram

1. Objectives

After completing this session, students will be able to:

- Write main functional requirements using clear action-oriented statements.
- Identify actors, system boundary and main use cases from a business scenario.
- Draw a use case diagram with suitable include/extend relationships.
- Prepare brief use case descriptions for selected main use cases

2. Scenario: University Internship Registration and Tracking

At a university, internship activities are currently managed through emails, paper forms, and separate Excel files. Students must find internship opportunities, ask lecturers or program coordinators for approval, and submit related documents manually. Staff members then record student internship information in spreadsheets and send updates through email or messaging applications.

This manual process causes several problems. Students do not always know whether their internship request has been approved, rejected, or is still being reviewed. Different staff members may record information differently, which leads to missing data, repeated data entry, and inconsistent tracking. Lecturers and coordinators also spend significant time answering repeated questions from students about approval status, company confirmation, and required documents. In addition, management cannot easily generate reports about internship participation, approved companies, or student progress.

To improve this situation, the university wants a simple computerized system that allows internship information to be recorded and tracked in a more organized way. The system should help staff manage student and company information,

support internship registration and approval, and allow students to track their request status.

System users:

- **Staff:** manage student, company, and internship information; record applications; update status; view reports.
- **Student:** request internship registration, upload required documents, and check application status.
- **Program Coordinator:** review and approve or reject internship requests.

Main processes:

- Manage Student Information
- Manage Company Information
- Register Student for Internship
- Review and Approve Internship Request
- Generate Internship Report

3. Requirements

Complete the following tasks.

3.1. Information Gathering

3.1.1. Identify Information Needs (10 pts)

List at least 5 types of information that are missing and need to be collected.

- Current average processing time for internship requests.
- List of companies currently partnering with the university for internships, and their accepted intake per semester.
- Specific documents required for an internship request (e.g., CV, company acceptance letter, insurance confirmation).
- The data fields currently recorded by staff in the Excel files.
- Number of internship requests handled per semester and the periods with the highest volume .
- The reports management currently needs.

3.1.2. Select Information Gathering Techniques (15 pts)

Complete the table below:

Information Needed	Source	Technique	Reason
Student difficulties and expectations when registering for an internship	Students	Questionnaire / Survey	Reaches many students quickly and reveals common

			patterns and expectations
Current approval workflow and reasons for delay	Program Coordinator	Interview	Provides detailed explanation of approval steps and the reasoning behind decisions
Data fields and required documents for internship applications	Existing Excel files and forms	Document Analysis	Existing spreadsheets and forms show the real data fields, formats, and rules already used
How staff actually record and update internship information day-to-day	Staff (Internship/Academic Office)	Observation	Reveals real working practice, including delays and manual workarounds not mentioned in interviews
Reporting needs for internship participation and approved companies	Management / Program Coordinator	Interview	Clarifies exactly what summary information decision-makers need from the new system

Choose from:

- Interview
- Questionnaire / Survey
- Observation
- Document Analysis

3.1.3. Design Interview Questions (15 pts)

Choose ONE stakeholder (Student / Staff / Program Coordinator)

Selected stakeholder: Student

Write:

- 3 Open-ended questions
 - What difficulties have you faced when trying to register for an internship?

- How do you currently find out whether your internship request has been approved, rejected, or is still being reviewed?
- What information or features would help you manage your internship application more easily?
- 3 Closed questions
 - Have you ever missed an internship deadline because you did not receive a timely update? (Yes / No)
 - Did you submit your internship documents on paper, by email, or online? (Paper / Email / Online)
 - On average, how many times do you contact staff to ask about your request status? (0 / 1–2 / 3–5 / more than 5)

3.2. Requirements & Use Case

3.2.1. Problem Description and System Request (10 pts)

Summarize the problem (why the system is needed).

Internship activities are currently managed manually through email, paper forms, and separate Excel files. This causes several problems: students cannot easily know whether their request has been approved, rejected, or is still under review; staff record information inconsistently, which leads to missing data and repeated data entry; and lecturers/coordinators spend a lot of time answering repeated status questions instead of reviewing requests. Management also cannot easily generate reports about internship participation, approved companies, or student progress. A computerized system is needed to centralize internship information and make the registration and approval process more transparent and consistent.

- Write the goal
 - Develop a computerized Internship Registration and Tracking system that allows staff to manage student, company, and internship information centrally, supports online internship registration and approval, and allows students to track their request status in real time.
- Write 2–3 expected benefits.
 - Faster and more consistent internship approval process, since requests, documents, and decisions are stored and processed in one system.
 - Improved transparency for students, because they can check their request status themselves instead of repeatedly contacting staff or lecturers.
 - Accurate, centralized data that allows management to generate reliable reports on internship participation and approved companies.

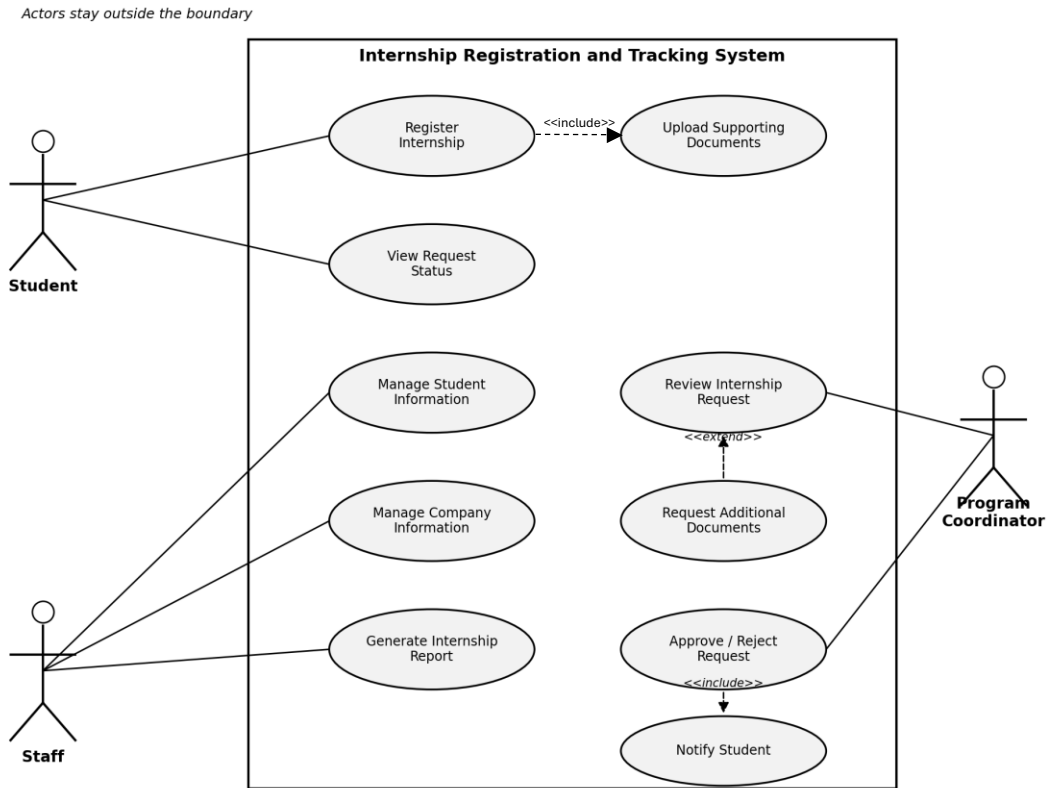
3.2.2. Functional Requirements (10 pts)

Write at least 05 main functional requirements from the scenario. Each requirement should start with “The system shall ...”.

No.	Main Functional Requirement	Related User/Actor
1	The system shall allow students to register for an internship online.	Student
2	The system shall allow students to upload required documents (e.g., CV, company acceptance letter) when submitting a request.	Student
3	The system shall allow students to view the current status of their internship request.	Student
4	The system shall allow staff to record and manage student and company information.	Staff
5	The system shall allow the program coordinator to review, approve, or reject internship requests.	Program Coordinator
6	The system shall notify students when their internship request status changes.	Student / System
7	The system shall allow staff to generate internship participation reports.	Staff

3.2.3. Use case diagram (20 pts)

- List of main use cases
 - Register Internship (Student)
 - Upload Supporting Documents – extend of Register Internship (Student)
 - View Request Status (Student)
 - Manage Student Information (Staff)
 - Manage Company Information (Staff)
 - Review Internship Request (Program Coordinator)
 - Request Additional Documents – extend of Review Internship Request (Program Coordinator)
 - Approve/Reject Request (Program Coordinator)
 - Notify Student – include of Approve/Reject Request (system)
 - Generate Internship Report (Staff)
- Draw a use case diagram, include:
 - Actors: Staff, Student
 - The main functions as use cases
 - Show system boundary and its name



3.2.4. Prepare brief use case descriptions (10 pts)

Choose 2 important use cases and write a brief description for each.

Use Case Name	Primary Actor	Trigger	Main Success Scenario	Alternative / Exception
Register Internship	Student	Student wants to apply for an internship at a company.	1) Student logs in. 2) Student selects an available internship/company. 3) Student enters required information. 4) Student uploads supporting documents. 5) Student submits the request. 6) System saves the request with "Submitted" status and notifies the coordinator.	If required documents are missing, the system prompts the student to upload them before the request can be completed.

Approve/Reject Request	Program Coordinator	A new internship request is waiting for review.	1) Coordinator opens the list of pending requests. 2) Coordinator selects a request. 3) Coordinator reviews student information and documents. 4) Coordinator approves or rejects the request. 5) System updates the request status and notifies the student.	If documents are incomplete, the coordinator selects "Request Additional Documents" and the system asks the student to provide the missing information before the request can be finalized.
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3.2.5. Check consistency (10 pts)

Map each selected requirement to at least one use case:

Requirement No.	Related Use Case	Consistency Note
1	Register Internship	Directly supports FR1 – allows the student to submit an internship registration online.
2	Register Internship (includes/extends Upload Supporting Documents)	Supports FR2 – documents are uploaded as part of the registration use case.
3	View Request Status	Directly supports FR3 – student can check the status of the request at any time.
4	Manage Student Information	Directly supports FR4 – staff maintain student records in the system.
5	Approve/Reject Request (extends Review Internship Request)	Supports FR5 – coordinator reviews and decides on internship requests.
6	Notify Student (included in Approve/Reject Request)	Supports FR6 – student is automatically notified whenever request status changes.
7	Generate Internship Report	Directly supports FR7 – staff can produce internship participation reports.

4. Submission

- Submit one PDF report.
- File name format: CSW309_Lab02_StudentNo_StudentName.pdf
- The submission should be written clearly and organized by task number.